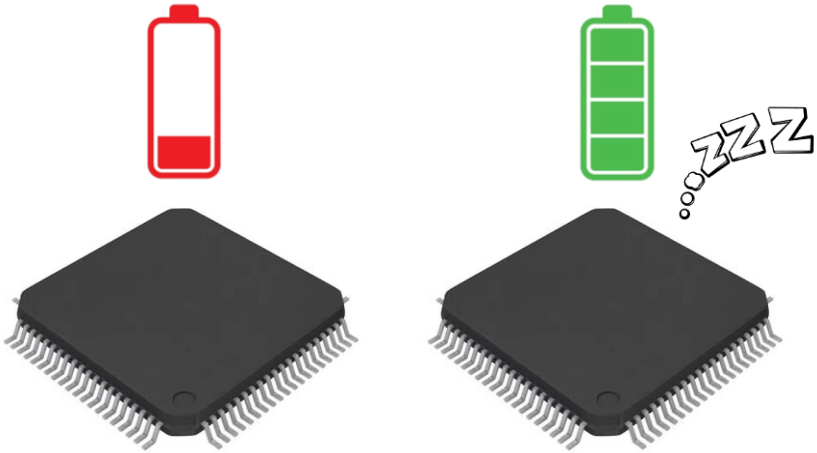
Why Sleep Modes Matter in MCUs



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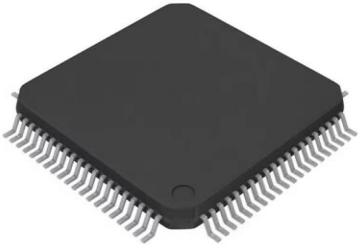
- Many devices like smartwatches, sensors, and IoT gadgets run on batteries.
- When the MCU stays active, even during idle periods, battery life drops quickly.
- Sleep mode is an efficient way to save power during idle periods.



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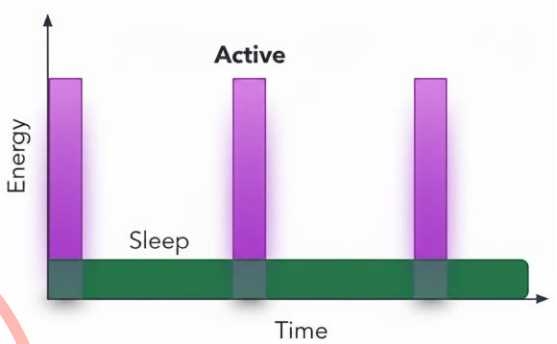
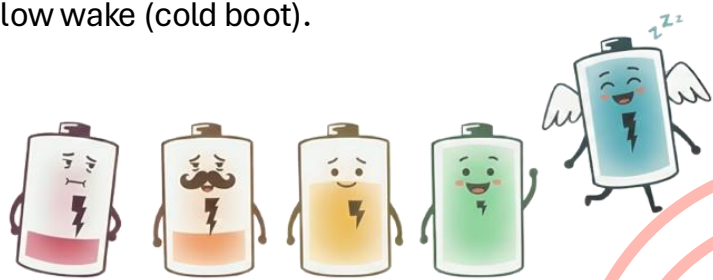
What Are Sleep Modes?

- Sleep modes let the MCU pause or slow down.
- Essential parts stay active (timers, interrupts).
- MCU wakes automatically on events.
- Saves energy without fully turning off.



Types of Sleep Modes

- **Active**: MCU fully running, highest power use.
- **Light Sleep:** CPU paused, peripherals active, quick wake.
- **Standby:** CPU off, some memory retained, low power, faster wake
- **Deep Sleep:** Most components off, only wake sources active, very low power
- **Hibernation:** Almost off, longest battery life, very slow wake (cold boot).



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How Sleep Modes Work

1. In sleep mode, the CPU slows down or stops while selected peripherals remain active.

2. The MCU wakes on interrupts or timed events.

3. Most of the time is spent sleeping, with short active periods to do tasks.

4. This reduces average power consumption without losing functionality.

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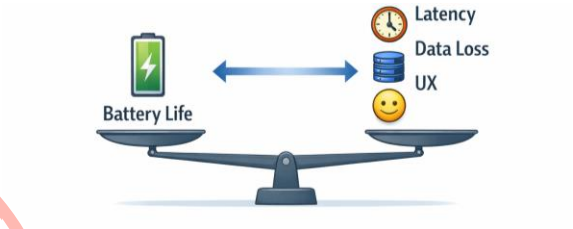
Typical Power States

Mode	Current Draw	Wake-up Time	Use Case
Active	mA	Instant	Continuous processing
Light Sleep	100s µA	µs	Event driven tasks needing fast response
Standby	1–10 µA	ms	Fast wake with state retention
Deep Sleep	nA–µA	ms–s	Infrequent periodic sensing

Trade-off = power vs responsiveness

The Real Design Problem

- The real problem in not unawareness to use deep sleep.
- It's balancing:
- Battery life ↔ Latency ↔ Data loss ↔ UX



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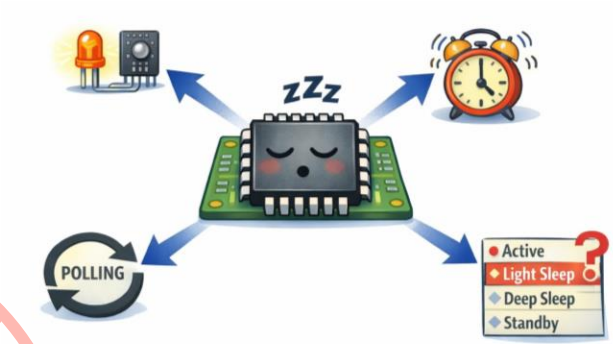
Common Mistakes We See

- Peripherals left running
- Too frequent wakeups
- Polling instead of interrupts
- Wrong sleep state choice

Result: 2×–10× wasted energy.



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